

Enhancing Memory Retention through the Strategic Timing of VR Transitions

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Abstract

This study investigates the impact of VR transition spaces on memory retention through two experiments (N=20 preliminary, N=40 main). While transition spaces are increasingly used in VR field to enhance immersion and presence, how they might enhance and contribute to learning outcomes remains unclear. Our goal was to advance the design of VR learning environments by investigating how strategically-placed transition spaces affect memory retention. Using a between-subjects design with four conditions (Transition-In, Transition-Out, Transition In-Out, and Control), we examined how transition timing affects memory construction and retention. Results revealed a noticeable variation between transition conditions, with Transition-In condition showing superior immediate retention compared to other conditions. However, this advantage diminished in 7-day follow-up testing, while other conditions demonstrated stable retention. These findings advance VR learning design theory by suggesting timing as a crucial factor in transition space effectiveness, providing insight into designing better digital learning experiences.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI; Interaction design; Empirical studies in interaction design; • **Human-center computing** → Human computer interaction (HCI); Interaction paradigms; Virtual reality.

Keywords

VR Transition, Transition Spaces, Memory Retention, Digital Learning, VR Learning

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1 Introduction

Transitions support cognitive processes by making sense of new context as we move from one state to another [10]. In virtual reality

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(VR), where immersive experiences blur boundaries between physical and virtual worlds, the concept of transition spaces is evolving and gaining renewed attention. While researchers have explored presence and immersion in VR, the specific role of transition spaces in memory retention remains under-explored.

1.1 Transition Spaces

Transition spaces have evolved across physical, digital, and virtual environments, each revealing fundamental principles about human experience. Physical spaces like lobbies facilitate both movement and psychological transitions, while digital transitions serve dual purposes of maintaining engagement and managing technical processes [1]. In VR, these spaces help preserve immersion during navigation, with studies exploring presence enhancement through preambles [16], narratives [19], and the effectiveness of gradual transitions [18].

Kim et al. [4] examined transition spaces across physical, digital, and virtual environments, identifying two fundamental types: transition-focused spaces, which emphasize spatial relationships, and experience-focused spaces, which center on interactive activities. However, the role of transition spaces in supporting attention, learning, and engagement has not been experimentally investigated.

1.2 Attributes of Transition Spaces

Two attributes support memory formation in transition spaces: physical interaction and spatiality. These properties can be understood through two theoretical frameworks: Embodied Cognition and the Method of Loci (MoL).

Embodied cognition emphasizes the role of physical interactions within environments in shaping cognitive processes. Wilson [20] posits that cognition is deeply grounded in bodily interactions with the environment. Within VR, studies have demonstrated that physical interactions aligned with learning content significantly enhance both immediate and long-term memory recall [2, 3]. Similarly, Ratcliffe and Tokarchuk [11] found that participants who engaged physically with virtual objects exhibited improved learning outcomes without increasing cognitive load, which refers to how much information that working memory is processing at any given time [14]. Based on this theory, the interactive elements of our transition spaces were specifically designed to be manipulated like physical objects.

The Method of Loci (MoL) offers a complementary perspective by emphasizing the role of spatiality in enhancing memory recall. This classical memorization technique leverages spatial environments by associating information with specific locations [22]. Early research by Roediger [13] confirmed the effectiveness of MoL in

enhancing memory retention. More recently, Krokos et al. [6] and Reggente et al. [12] found that users in VR-based MoL systems demonstrated superior recall compared to traditional memorization methods, indicating the benefits of immersive environments. More recently, Moll and Sykes [8] showed that VR implementations achieved approximately a 20.4% improvement over conventional techniques, highlighting the potential of well-designed VR environments as effective memory anchors. While MoL traditionally relies on imagined spaces to create cognitive waypoints, VR transition spaces in our study may offer tangible, navigable environments. By incorporating spatial and experiential cues within transition spaces, VR environments may serve as tools that enhance memory recall more effectively than imagined spaces.

1.3 Timing of Transitions

Task switching theory provides a crucial framework for understanding how transition timing affects memory retention [17]. Transitions between tasks or environments often result in a “switching cost”—the additional cognitive effort required to adjust to new tasks [15]. Two key cognitive processes contribute to this phenomenon: task-set reconfiguration and task-set inertia.

Task-set reconfiguration refers to the cognitive delays which occur when preparing for a new task. The cue-stimulus interval (CSI)—the time between a cue and the task—plays a crucial role here. Users can prepare more effectively for the switch when the CSI is long enough, leading to reduced switching costs. However, if the CSI is too short, the lack of preparation time leads to higher switching costs [14].

Task-set inertia, on the other hand, suggests switching costs result from lingering cognitive engagement with the previous task. This can slow responses and lead to more errors when switching to a new task [15, 21]. However, in cases where the same task is repeated, this inertia can help by keeping the mind focused on the same task, reducing the mental effort required to switch [5, 7].

Drawing from task-switching theory, the timing of transition spaces in VR can influence cognitive processes differently, either by preparing users for an upcoming learning through adequate CSI or by helping them consolidate information through managed task-set inertia. This theoretical framework offers guidance for how VR transition timing might affect memory retention.

2 Experiments

We conducted preliminary (N=20) and main experiments (N=40) in a university lab using Meta Quest 2 headsets. Both experiments gained IRB approval that includes conducting experiments, analyzing the results, collecting a survey, and conducting in-depth interviews that students voluntarily joined. The prototype environments were created using Unity 22.3.9 and were based on the Unity XR Interaction Toolkit.

2.1 Preliminary Study Insights

Our main experiment design was informed by a preliminary study (N=20) that explored both transition types and timing in VR learning environments. This initial investigation examined whether transition spaces enhance memory retention, and how their timing (pre- or post-learning) affects cognitive processing.

The study employed a between-subjects design with participants randomly assigned to five conditions: two types of transition spaces (hallways or lobbies) placed either before or after learning content, and a control condition with no transitions. The learning environment featured unfamiliar indigenous art and historical knowledge to minimize prior knowledge effects and increase the likelihood that our participants (graduate design students) would be motivated to learn.

The most valuable insight from this preliminary investigation was the contrast between objective performance measures and subjective experiences. While quantitative memory retention scores showed modest differences between conditions, the qualitative feedback revealed a critical pattern: participants who experienced pre-learning transitions reported these spaces helped prepare them mentally for learning. As one participant noted, “The first room after training gave me a sense of what I was about to learn, which made the content easier to absorb.” Conversely, participants experiencing post-learning transitions often questioned the purpose of the transition spaces.

These preliminary findings, particularly the divergence between quantitative performance and qualitative perception regarding transition timing, directly shaped the focus of our main experiment. Rather than continuing to compare transition types (hallways vs. lobbies), we chose to focus exclusively on timing effects with an expanded sample size, hypothesizing that the subjective perception differences might manifest more clearly in objective performance with a refined design and larger sample.

2.2 Main Experiment

Building on insights from our preliminary study, we designed a more comprehensive investigation focusing on the placement of transition spaces and their impact on both immediate and long-term memory retention. In addition to the conditions from the preliminary study—Transition-In (transition before learning phase) and Transition-Out (transition after learning phase)—we added a third condition: Transition-In-Out, where participants experience transitions both before and after the learning phase. We selected the Korean alphabet (Hangul) as our learning content, for its systematic geometric structure and novelty to our non-Korean speaking participants. We hypothesized that participants experiencing any form of transition space would demonstrate higher memory retention compared to those in the control condition (H1), and that participants experiencing transitions both before and after learning would achieve the highest memory retention scores among all conditions (H2). For long-term memory, we hypothesized that the pattern of memory retention observed in immediate testing would persist after seven days (H3).

2.2.1 Participants. After excluding participants who are familiar with the learning content and who experienced technical difficulties, cyber sickness, or procedural interruptions, our final sample comprised forty graduate design students (26 female, 14 male; age $\bar{x}$ = 28.73, Mdn = 27). The study employed a between-subjects design where participants were randomly assigned to one of four conditions: Transition-In: transition before learning (n = 10), Transition-Out: transition after learning (n = 10), Transition In-Out: transition

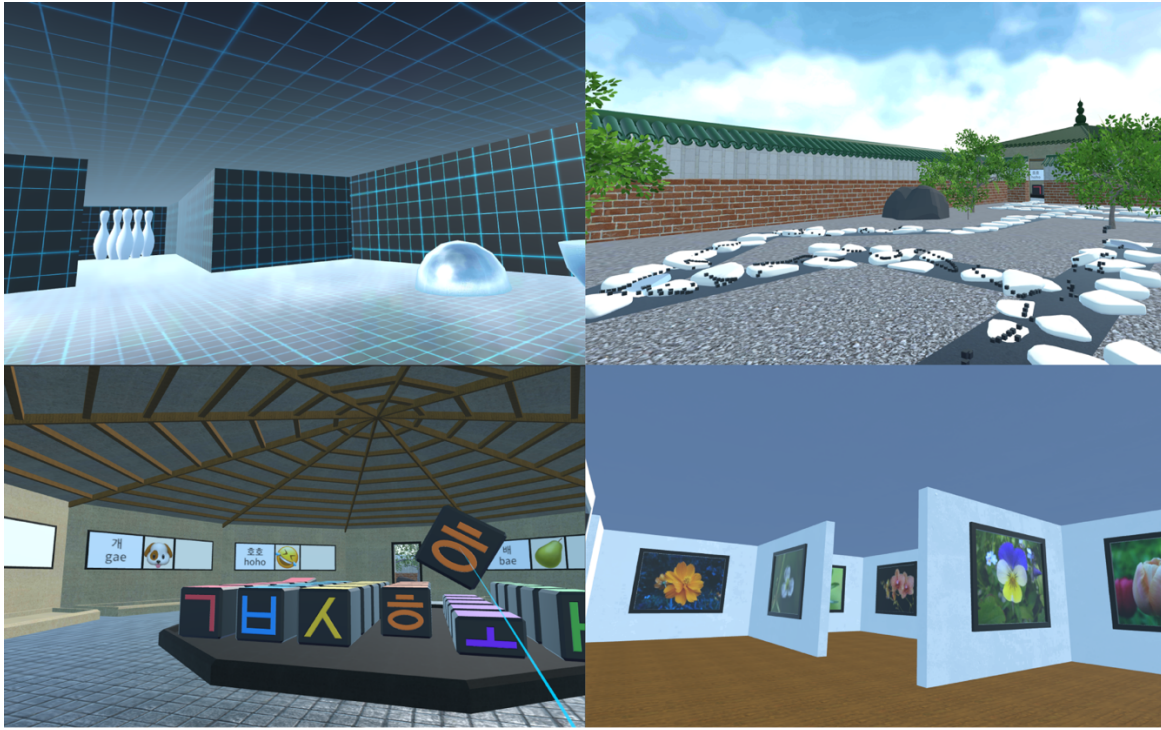


Figure 1: Top left: Gaming space, top right: transition space, bottom left: learning space, bottom right: contemplating space

Table 1: Experiment Sequence for Each Condition

Condition	Gaming	First transition	Learning	Second transition	Contemplating
Transition-In	Bowling alley	Courtyard with “Sogae” pathway	Hangul learning space	X	Flower photo gallery
Transition-Out	Bowling alley	X	Hangul learning space	Courtyard with “Bae” pathway	Flower photo gallery
Transition In-Out	Bowling alley	Courtyard with “Sogae” pathway	Hangul learning space	Courtyard with “Bae” pathway	Flower photo gallery
Control	Bowling alley	X	Hangul learning space	X	Flower photo gallery

both before and after learning ($n = 10$), Control: no transitions at all ($n = 10$).

2.2.2 Prototypes and Experiment Design. Based on insights from embodied cognition theory and Method of Loci, we designed each virtual environment in our study to offer unique experiences in order to maintain clear transitions between them and aid memorability (Figure 1).

In the Gaming Space, a bowling alley introduced participants to VR interactions through an engaging bowling activity, intentionally differing from subsequent spaces to highlight transitions while teaching essential mechanics. In the Transition Space, a courtyard featured interactive pathways shaped as Hangul words (“소개” / “Sogae” meaning “introduction” or “배” / “Bae” meaning “pear”), encouraging engagement and indirectly reinforcing learning. This design allowed participants to interact with the space

by creating distinctive and spatial markers. The Learning Space environment drew inspiration from traditional Korean *Hanok* architecture, and featured interactive 3D Hangul letter blocks (ㄱ, ㅍ, ㅅ, ㅊ, ㅌ, ㅍ) for forming eight target words. Each word appeared with pronunciation guides, emoji representations, and drawing canvases. Lastly, the Contemplating Space provided a decompression phase as a simple gallery environment featuring images of 21 diverse flower specimens, designed to provide a pause before assessment.

The experimental procedure guided participants through a series of distinct virtual environments. Table 1 provides an overview of the experiment sequence for each condition and Figure 2 shows a diagram of the experiment design. We designed our experimental sequence based on task switching theory concepts. The Transition-In condition examined whether a pre-learning transition served as

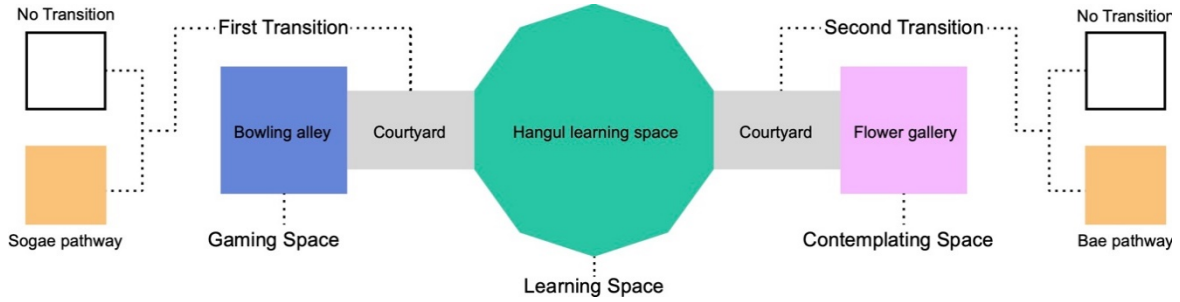


Figure 2: Diagram of VR Experiment Design

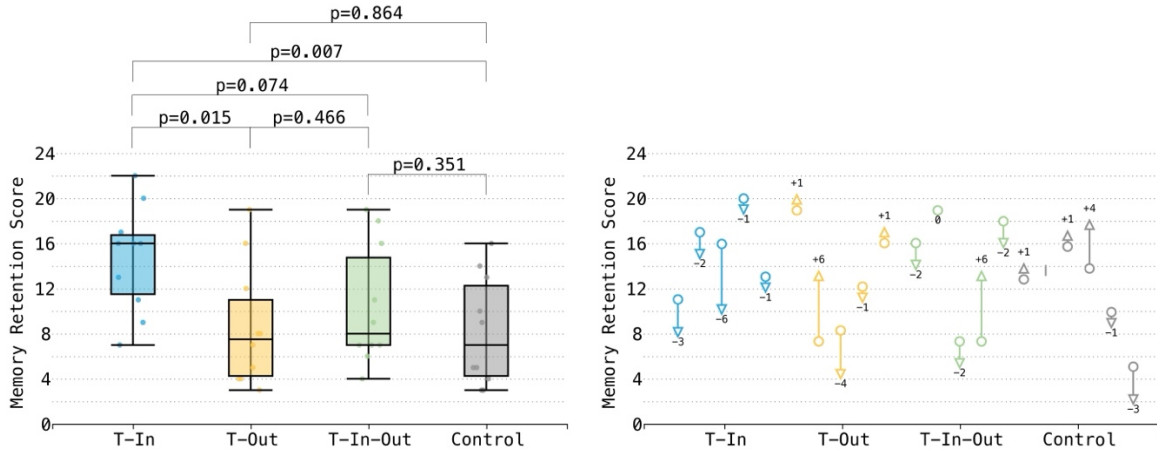


Figure 3: Left: Immediate Memory Retention Score by Condition, Right: Immediate vs. Long-term

an effective cue-stimulus interval, facilitating task-set reconfiguration before learning. The Transition-Out condition investigated whether a post-learning transition could support memory consolidation through managed task-set inertia. The Transition-In-Out condition combined both mechanisms to test their potential cumulative effects.

2.2.3 Learning Assessment. To assess retention, we developed a 24-point evaluation system, testing letter-sound correspondence (6 points) and word recognition and meaning (18 points). Participants completed this assessment at two times: immediately following the VR session and seven days later. While all forty participants completed the immediate assessment, a subset of twenty participants (five from each condition) returned to complete the delayed assessment. This follow-up testing used identical assessment materials to allow direct comparison with immediate retention scores.

3 Results

3.1 Memory Retention Scores

For all statistical analysis, we made use of the OriginPro software package. Before conducting the main analyses, we examined the data for violations of statistical assumptions. Shapiro-Wilk tests indicated normal distributions for all conditions (Transition-In: $W = 0.94$, $p = .52$; Transition-Out: $W = 0.96$, $p = .78$; Transition-In-Out: $W = 0.95$, $p = .67$; Control: $W = 0.93$, $p = .45$).

For H1, which predicted higher memory retention with transition spaces compared to control, one-way ANOVA revealed a meaningful effect of transition conditions ($F(3,36) = 3.39$, $p = .028$, partial $\eta^2 = .22$). Transition-In ($\bar{x} = 14.7$, $SD = 4.72$) outperformed the Control ($\bar{x} = 8.2$, $SD = 4.87$) (mean difference = 6.5, $p = .007$), but other transition conditions did not show significant improvements over the Control (all $p > .05$).

Contrary to H2, which predicted highest retention scores for the combined transition (Transition-In-Out), our results showed that the Transition-In-Out condition demonstrated only moderate retention ($\bar{x} = 10.4$, $SD = 5.38$), lower than the Transition-In alone. This suggests that additional transitions may not enhance, and might even diminish, memory formation benefits.

For H3, which predicted persistence of immediate retention patterns over seven days, our data revealed different patterns among the subset of twenty participants ($n = 5$ per condition). Due to the reduced sample size, we report descriptive statistics rather than inferential statistics. The Transition-In condition showed a marked decline in performance (mean decrease = 2.6 points, $SD = 1.8$), contrasting with their superior immediate retention. Other conditions maintained stable retention patterns with minimal changes (all mean changes < 0.5 points): Transition-Out ($SD = 1.2$), Transition-In-Out ($SD = 1.4$), and Control ($SD = 1.1$). While the Transition-In condition at immediate assessment showed advantage over the

Control, these benefits diminished over the retention period. However, given the small sample size and numerous external factors that could have influenced retention over the seven-day interval, these observations should be interpreted with caution.

The root mean square error ($RMSE = 5.11$) and coefficient of variation ($CV = 0.487$) indicated moderate variability across conditions. Post-hoc Tukey's HSD tests showed meaningful differences between specific conditions. The Transition-In condition outperformed the Control (mean difference = 6.5, $p = .035$, 95% CI [-12.65, -0.35]). A marginal advantage was found for the Transition-In condition over Transition-Out condition (mean difference = 6.1, $p = .053$, 95% CI [-12.25, 0.05]). Our findings demonstrate that pre-learning transitions enhanced immediate memory retention compared to the Control condition, though this advantage decreased over time.

3.2 In-depth Interview

Following the seven-day retention assessment, in-depth interviews provided insights regarding transition spaces and learning experiences. Several participants highlighted the preparatory benefits of transition spaces, with T-Out 7 and T-In-Out 7 noting that the courtyard with the Hangul word-shaped pathway helped them prepare for Hangul learning, not only by improving their familiarity with VR interactions and navigation, but also by setting their expectations for upcoming learning content. These comments suggest that transition spaces may primarily function as cognitive preparation tools, rather than memory reinforcement mechanisms.

The VR learning environment itself received positive feedback across conditions. Participants particularly valued the multimodal learning approach, with No-T 12 describing the Learning Space as very memorable due to its combination of emoji associations and building blocks. Many comments emphasized that interactive elements were helpful (noted by T-Out 5 and T-Out 7 as much better than 2D learning), indicating that the core learning environment design effectively supported initial learning, regardless of transition conditions.

Despite expressed interest from some participants (T-In 1, T-In 10, and T-In-Out 4), no participants self-reported pursuing further Hangul study after the experiment. These findings collectively suggest that while transition spaces, particularly pre-learning transitions, can enhance immediate memory retention, their role in supporting long-term retention may be more complex and dependent on additional factors outside of the initial learning experience.

4 Discussion and limitations

Our findings contribute to the understanding of how transition spaces in VR environments affect memory retention, with important implications for both theory and design practice. The differential impact of transition timing that we observed reveals several key considerations for future VR learning environments.

The superior performance of the Transition-In condition aligns with our insights from task-set reconfiguration theory, suggesting that pre-learning transitions serve as effective preparation periods. This finding points to a potential design principle for building VR

learning environments: transition spaces should be strategically placed before complex or unfamiliar learning content. Future research could explore the optimal duration and characteristics of these preparatory periods.

The relatively lower performance of the Transition-Out condition challenges assumptions about post-learning reflection periods [9]. While existing literature suggests that time for reflection helps learners maintain focus on the same task, our findings indicate that post-learning transitions may not effectively support memory construction. We believe that this observation may be due to task-set inertia, with the transition space perhaps interrupting natural memory consolidation processes.

The moderate performance of the Transition-In-Out condition suggests that additional transitions may create unnecessary complexity rather than complementary benefits. This raises important questions about cognitive load in VR learning environments that merit further investigation. Researchers should explore whether this threshold varies based on interaction complexity, learner characteristics, or specific learning objectives. Future work should investigate optimal transition frequency based on content difficulty and learner characteristics, potentially exploring adaptive transition spaces that respond to learner performance and behaviors.

The unexpected decline in the Transition-In condition's performance over seven days, compared to the stability of other conditions' scores, suggests that different mechanisms may influence immediate versus long-term memory formation in VR learning. However, given the limited sample size for the delayed assessment and mediating factors during the seven-day gap, this interpretation should be approached with caution. Nevertheless, these patterns raise important questions for future research on how transition-enhanced learning effects might be supported over time. Longitudinal studies that involve variable transitions across multiple lessons and time periods would clarify these effects.

The findings of this study are subject to certain limitations. First, the focus on graduate design students limits generalizability to other populations. The reduced sample size for the follow-up assessment constrains our ability to draw robust conclusions about long-term retention effects. Another limitation lies in the content itself—Hangul letters and words—which may have influenced participants' engagement due to the unfamiliarity of the material.

Our current analysis treated transition conditions as distinct categories, and future studies with larger samples could explore a factorial analysis examining other characteristics of transitions such as their durations, relatedness with learning content, accessibility dimensions, and variability over time. Such an approach might reveal whether transitions combine additively to influence memory retention, providing deeper insights into the mechanisms of transition timing in VR learning environments.

5 Conclusion

Our findings suggest several critical considerations for the design of virtual reality learning environments. First, the timing of transitions appears more crucial than their presence, with pre-learning transitions showing promise for supporting immediate memory retention. Virtual reality educational designers should prioritize the strategic temporal and spatial placement of transitions, rather

than unsystematically incorporating them throughout learning experiences.

Additionally, the relative ineffectiveness of post-learning transitions suggests the need to reconsider how we support post-learning consolidation in virtual reality environments. Traditional supportive approaches, such as reflection periods after extensive learning content, may require substantial rethinking for immersive contexts; particularly when considering the unique cognitive demands of virtual reality interaction.

While our study demonstrates the potential of carefully timed transition spaces in VR learning environments, it also reveals the complexity of transition sequence effects, emphasizing the need to consider not just the presence of transitions, but their timing, frequency, and relationship with learning experiences.

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